

A tritone-centered self-duality: the set class 6-30 and its order-12 neo-Riemannian dual

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Abstract

Stravinsky’s *Petrushka* chord stacks two major triads a tritone apart. We follow that tritone into pure group theory and back. Over the chromatic universe $\mathbb{Z}_{12}$, the T/I group acts regularly on the orbit of a pitch-class *set* class — admitting a simply-transitive dual contextual group of reduced order in Lewin’s sense — precisely when the class’s T/I -stabilizer is *normal*; among order-2 symmetries this singles out the center $\{T_0, T_6\}$, the tritone. The unique nontrivial class in $\mathbb{Z}_{12}$ with that symmetry is Forte **6-30** = (013679), the set class of the *Petrushka* chord; its orbit has size 12 and its dual is dihedral of order 12 ($\cong D_6$). We give the characterization with proof, identify the instance, enumerate the phenomenon across universes (the count is $A032239(n/2)$, asymmetric two-color bracelets), and **machine-check** in Lean 4 (sorry-free, axiom-clean) both the background duality engine of Crans–Fiore–Satyendra and the order-12 regular dihedral structure of 6-30’s dual. The mathematics is classical; the contributions are the specific characterization, the 6-30 instance with its first machine-checked dual, and the enumeration.

1 The tritone, from Stravinsky to a group

In the second tableau of *Petrushka* (1911), Stravinsky sounds a C-major triad against an F♯-major triad — two major triads a *tritone* apart. As pitch classes (with $C = 0$) this “*Petrushka* chord” is $\{0, 1, 4, 6, 7, 10\}$ [b audio], Forte set class **6-30**, a six-note subset of the octatonic collection that the Russian tradition handed Stravinsky through Rimsky-Korsakov [11, 12, 13].¹

P₀. Why should this collision — two triads a tritone apart — feel so self-contained? What is special, structurally, about the tritone?

This note follows that question into the algebra of transformations and back. The neo-Riemannian tradition (Lewin [4]; Crans–Fiore–Satyendra [1]) reads chords not by their notes but by the group of operations relating them, and pairs each transformation group with a *dual*. We ask what that duality

*Prepared with the assistance of an AI system (Claude, Anthropic). The mathematics is classical; every claim, scope statement, and limitation is the author’s responsibility. The boundary of the contribution is stated plainly in Section 8, and the Lean 4 kernel—not the assistant—is what certifies the formalized parts.

¹The octatonic reading of Stravinsky is the mainstream analytical tradition (Berger, van den Toorn, Taruskin); for the dissenting view that it is over-applied, see Tymoczko [14]. We use “*Petrushka* chord” as the name of the sonority, after the ballet, not as a claim about any single coiner.

does for a *symmetric* chord like Petrushka's, prove exactly when a reduced dual exists, and find that the answer is governed throughout by the tritone — which turns out to be, at once, the musical heart of the chord and the center of the group. Every claim about 6-30 below is machine-checked in Lean 4.

2 Setup: triads, the T/I group, and its dual

To follow the tritone we first need the cast: the chords, the group that moves them, and Lewin's notion of a *dual* group. Pitch classes are $\mathbb{Z}_{12}$. The T/I group is the dihedral group D_{12} of order 24: transpositions $T_k : x \mapsto x + k$ and inversions $I_k : x \mapsto -x + k$. We identify it with Mathlib's `DihedralGroup 12` ($ri = T_i$, srj an inversion). A *triad* is a pair $(\text{root}, \text{quality}) \in \mathbb{Z}_{12} \times \{\pm\}$ (major $\{r, r+4, r+7\}$ or minor $\{r, r+3, r+7\}$); there are 24, and T/I acts on them simply transitively — the left regular representation of D_{12} . The neo-Riemannian operations P, L, R generate the group dual to T/I on the 24 triads: two subgroups of $\text{Sym}(\Omega)$ are *dual* (Lewin) when each is the centralizer of the other and both act simply transitively [1].

P₁. The triad is asymmetric, so its orbit has 24 forms and the duality is of order 24. What happens to the dual when the object is a symmetric chord, whose orbit is smaller?

The duality is the regular case of the classical centralizer theorem (the centralizer of a regular permutation group is its opposite regular representation; Dixon–Mortimer [2] §4.2, Theorem 4.2A; Cameron [3]). We machine-check the engine and the $T/I \leftrightarrow PLR$ instance in Lean (Table 3, namespace `Duality`): the reusable Wielandt regular-case lemma `centralizing_fixedPoint_eq_one`, the bound `card_PLRgrp = 24`, and the duality itself `centralizer_eq_PLR`: $\text{centralizer}(T/I) = \langle P, L, R \rangle$.

3 When does a symmetric chord keep a dual?

A symmetric class has a nontrivial T/I -stabilizer $H = \text{Stab}(A)$, so its orbit $O = T/I \cdot A \cong D_{12}/H$ has size $24/|H| < 24$, and the T/I -action on O is transitive but no longer free. Does O still carry a simply-transitive dual?

Proposition 1 (Reduced duality). *Let A be a pitch-class set with T/I -stabilizer $H = \text{Stab}(A) \leq D_{12}$ and orbit $O = T/I \cdot A$. The centralizer of the T/I -image in $\text{Sym}(O)$ acts simply transitively — a reduced-order dual group in Lewin's sense — if and only if H is normal in D_{12} , and then the dual has order $|O| = 24/|H|$.*

Proof. D_{12} acts transitively on $O \cong D_{12}/H$ with point-stabilizers the conjugates of H . For a transitive action of a group G with point-stabilizer H , the centralizer C of the image in $\text{Sym}(O)$ is isomorphic to $N_G(H)/H$ and is semiregular; C is transitive — equivalently regular, equivalently the dual is simply transitive — iff $N_G(H) = G$, i.e. iff $H \trianglelefteq G$ ([2] §4.2; [3]). When $H \trianglelefteq G$, $|C| = |G/H| = |O|$. $\square$

P₂. Which symmetry is that, musically? Is there a chord whose stabilizer is exactly the tritone — and is it unique?

The normal subgroups of D_{12} contained in the rotation group are exactly the $\langle T_d \rangle$; reflections (inversions) generate non-normal subgroups. Restricting to an *order-2* symmetry isolates the tritone:

Lemma 1 (The tritone is the central involution). *T_6 is the unique element of order 2 in the rotation group $\mathbb{Z}_{12}$ (equivalently, 6 is the unique nonzero fixed point of $x \mapsto -x$), and $\{T_0, T_6\} = Z(D_{12})$ is*

the unique normal order-2 subgroup of D_{12} . A single inversion $\langle I_j \rangle$ is an order-2 subgroup but is not normal.

So among set classes with an order-2 symmetry, Proposition 1 gives a reduced (order-12) simply-transitive dual *iff* the symmetry is the tritone T_6 . A non-central order-2 symmetry (an inversion) gives no dual. The 222-class brute-force sweep (`sage/duality_sweep.sage`) corroborates the general statement with zero exceptions; the proof above is what makes it a theorem rather than an observation.

4 6-30: the answer, and it is the Petrushka chord

Lemma 1 told us where to look: a chord stabilized by the tritone T_6 and nothing else. In $\mathbb{Z}_{12}$ there is exactly one, and we have already met it.

Theorem 1 (The 6-30 instance). *The unique nontrivial set class in $\mathbb{Z}_{12}$ whose T/I -stabilizer is exactly $\{T_0, T_6\}$ is **6-30** = (013679). Its orbit has size 12, and its dual group is regular and dihedral of order 12 ($\cong D_6$).*

$A = (013679)$ is the union of the three *central pairs* (tritones) $\{0, 6\}, \{1, 7\}, \{3, 9\}$, so it is fixed by T_6 (a half-turn of the pitch-class clock, Figure 1) [b audio]; having no inversional symmetry, $\text{Stab}(A) = \{T_0, T_6\}$ exactly. Its inversion $(0, 1, 4, 6, 7, 10)$ is the Petrushka chord itself (the 6-30A/B pair). **The aha of P_0 :** the tritone bitonality Stravinsky reached for by ear *is* the central symmetry T_6 , and that central symmetry is exactly what endows 6-30 with its order-12 self-duality.

This entire theorem is machine-checked in Lean 4 (file `SixThirty.lean`, Table 4), sorry-free and axiom-clean: $\text{Stab}_T(A) = \{0, 6\}$ and no inversional symmetry (`stab_T`, `stab_I_empty`); orbit size 12 (`orbit_card`); the kernel of the D_{12} -action on the orbit is the center $\{T_0, T_6\}$ (`Ker_eq`), so the image is D_{12}/Z of order 12 (`card_TIgrp0`); the dual (centralizer on the orbit) has order 12 (`card_Cgrp`, built as the *opposite regular representation*) and acts transitively, hence regularly (`Cgrp_transitive`); and it carries the dihedral fingerprint (`dihedral_fingerprint`: an order-6 generator s , an involution $t \neq 1$ with $tst = s^{-1}$ and $st \neq ts$), pinning it as dihedral of order 12.

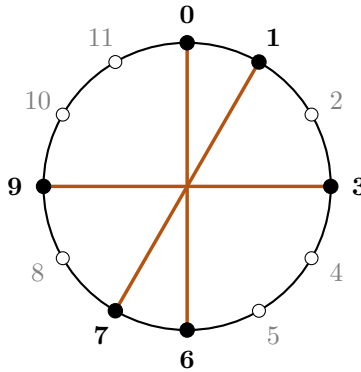


Figure 1: **6-30** = $\{0, 1, 3, 6, 7, 9\}$ on the pitch-class clock (filled nodes). It is the union of the three central pairs $\{0, 6\}, \{1, 7\}, \{3, 9\}$ (heavy tritone diameters), hence fixed by T_6 (a 180° rotation); with zero inversional symmetries it is *not* inversionally symmetric. Choosing three of the six tritone pairs, up to rotation/reflection, is the bracelet bijection of §5 (Figure 3).

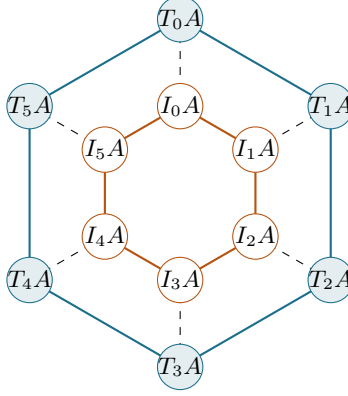


Figure 2: Cayley graph of the order-12 dual group D_6 acting simply transitively on the 12-element T/I -orbit of 6-30: the six transposition-forms $T_k A$ (outer, $T_k A = T_{k+6} A$ identified) and the six inversion-forms $I_k A$ (inner). Solid arrows are an order-6 generator (orientation reverses on the inner coset, as in any dihedral Cayley graph); dashed edges an order-2 generator. Regularity of this action (`Cgrp_transitive` + semiregularity) is what makes the dual a group of order 12.

5 How often does this happen?

P₃. Is $\mathbb{Z}_{12}$ special, or does a tritone-centered self-dual exist in every universe? How many are there?

For even n , count the set classes whose T/I -stabilizer is exactly the center $\{T_0, T_{n/2}\}$ (dual $\cong D_{n/2}$, orbit n), by ranging over unions of the $n/2$ central pairs $\{x, x + n/2\}$ (`sage/sixthirty.sage`):

n	8	10	12	14	16	18	20	24
count	0	0	1	2	6	14	30	127

Table 1: Tritone-centered self-dual classes per universe n : none below 12, unique at 12 (the class 6-30).

So $\mathbb{Z}_{12}$ is the smallest universe where the phenomenon exists, and there it is unique. The sequence 1, 2, 6, 14, 30, 62, 127 for $n = 12, 14, \dots, 24$ is **OEIS A032239** (“number of identity (asymmetric) bracelets of m beads, 2 colors”) at $m = n/2$. The bijection (Figure 3): a tritone-centered self-dual class is a choice of subset of the $n/2$ central pairs with trivial symmetry, i.e. an asymmetric two-color bracelet on $n/2$ beads. There is no simple closed form (the tempting $2^{m-5} - 2$ fits $m = 7..11$ then breaks at $m = 12$: $A032239(12) = 127 \neq 126$); the enumeration is a Burnside/Pólya specialization. What is ours is the musical characterization, the bracelet bijection, and the $\mathbb{Z}_{12}$ -minimality of 6-30.

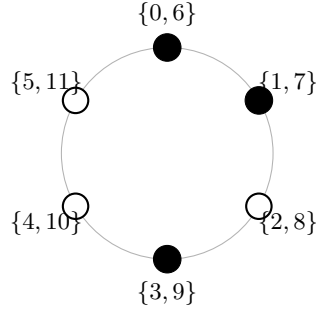


Figure 3: 6-30 as a two-color bracelet on $m = 6$ beads: bead i is the central pair $\{i, i + 6\}$, filled iff in the set. The pattern (filled at $\{0, 6\}, \{1, 7\}, \{3, 9\}$) has trivial rotation/reflection symmetry — an *asymmetric* bracelet, the object counted by A032239; 6-30 is the unique such pattern at $m = 6$.

6 What the dual is, musically

P₄. The dual is an abstract group of order 12. What is it for?

A simply transitive group action is exactly a *Generalized Interval System* in Lewin’s sense [4, p. 157]: given any two elements there is a *unique* operation carrying one to the other. So the dual $\cong D_6$ is a GIS on the twelve forms of the Petrushka chord [b audio] — Figure 2 is its map. Any form of the chord reaches any other by a unique dual move; the twelve Petrushka colors become a single, navigable harmonic space, the symmetric-chord analogue of the $P/L/R$ network for triads. Table 2 places the result in its family.

Hear it. Three sonifications accompany this note — the Petrushka chord (two triads a tritone apart), the T_6 invariance (6-30 and its tritone transpose share the same six pitch classes), and a traversal of all twelve forms (the dual-group orbit, passing through the Petrushka chord at I_1) — as editable MIDI and rendered WAV: <https://github.com/karlesmarin/music-math/tree/main/media>.

Object	Stabilizer	Orbit	Dual	Order
Triad (Crans–Fiore–Satyendra)	$\{T_0\}$	24	D_{12}	24
Hexatonic cycle (Berry–Fiore)	$\{T_0, T_4, T_8\}$ (aug.)	6	S_3	6
6-30 (this note)	$\{T_0, T_6\}$ (tritone)	12	D_6	12

Table 2: Three instances of the same centralizer-duality machine, by the symmetry of the object.

7 What is machine-checked

P₅. Can we trust all this?

Everything stated about the duality engine and about 6-30 is checked by the Lean 4 kernel (Tables 3–4); `#print axioms` returns exactly `[propxt, Classical.choice, Quot.sound]` with no `sorryAx` on every theorem. The enumeration (Table 1) and the named structure D_6 are additionally witnessed by re-runnable Sage/GAP scripts (Table 5).

Theorem	Statement
<code>centralizing_fixedPoint_eq_one</code>	Wielandt regular case (centralizer of a transitive group is semiregular).
<code>PLR_le_centralizer</code>	$\langle P, L, R \rangle \leq \text{centralizer}(T/I)$.
<code>card_PLRgrp</code>	$\text{Nat.card PLRgrp} = 24$.
<code>centralizer_eq_PLR</code>	$\text{centralizer}(T/I) = \langle P, L, R \rangle$ (the CFS duality).

Table 3: The background duality engine in `NeoRiemannian.lean` (namespace `Duality`), machine-checked (Brick 3).

Theorem	Statement
<code>stab_T, stab_I_empty</code>	$\text{Stab}_T(A) = \{0, 6\}$; no inversional symmetry.
<code>orbit_card</code>	T/I -orbit of A has 12 elements.
<code>Ker_eq</code>	kernel of the D_{12} -action on the orbit $= Z = \{T_0, T_6\}$.
<code>card_TIgrp0</code>	image on the orbit $= D_{12}/Z$, order 12.
<code>card_Cgrp</code>	the dual (centralizer on the orbit) has order 12 (opposite regular rep).
<code>Cgrp_transitive</code>	the dual is transitive, hence regular (simply transitive).
<code>dihedral_fingerprint</code>	order 12; s of order 6, $t^2 = 1$, $t \neq 1$, $tst = s^{-1}$, $st \neq ts$.

Table 4: 6-30’s order-12 regular dihedral dual in `SixThirty.lean`, machine-checked (Brick 19) — the first Lean proof of the structure of a symmetric chord’s contextual dual.

Script	What it witnesses
<code>centralizer_witness.sage</code>	GAP: $PLR == \text{centralizer}$ exactly; 6-30’s 12-orbit dual $\cong D_6$.
<code>sixthirty.sage</code>	6-30 detail (stab $\{T_0, T_6\}$, orbit 12, dual order 12) + counts per n .
<code>duality_sweep.sage</code>	all 222 $\mathbb{Z}_{12}$ classes: regular $\iff$ self-dual (0 exceptions).
<code>sc630_verify.sage</code>	$\text{IV} = [2, 2, 4, 2, 2, 3]$; stab $\{T_0, T_6\}$; not inversionally symmetric.

Table 5: Sage/GAP witnesses (Sage 10.8 + GAP), each re-runnable.

8 Novelty, scope, and open questions

Contribution, stated plainly. This is a formalization + characterization note, not new mathematics. The duality engine is the regular case of **Wielandt’s centralizer theorem** ([2] §4.2 Thm 4.2A; [3]); the contextual-group framework is Fiore–Satyendra/Fiore–Noll [7, 8, 9]. What is ours: (i) the reduced-duality characterization with proof (Proposition 1) and its tritone corollary (Lemma 1); (ii) the 6-30 instance with the *first machine-checked* contextual dual of a symmetric chord (Theorem 1, Brick 19); (iii) the enumeration and bracelet bijection. Mathlib has `DihedralGroup`, `IsPretransitive` and Cayley’s theorem, but no centralizer-of-regular-rep result; our `centralizing_fixedPoint_eq_one` and the opposite-regular construction in `SixThirty.lean` fill that gap as first formalizations.

Relation to Berry–Fiore. Berry–Fiore [5] prove that the PL -group of a hexatonic cycle is Lewin-dual to its T/I -stabilizer — the same machine. The objects differ (Table 2): theirs is the augmented-axis stabilizer $\{T_0, T_4, T_8\}$ with dual S_3 of order 6; ours is the tritone center $\{T_0, T_6\}$ with dual D_6 of order 12. “Tritone”, “octatonic”, “6-30” do not appear in their paper.

Relation to Peck. Peck [6] generalizes Lewin’s commuting groups to semiregular and diagonal actions with musical examples; we cite him for the machine. To our knowledge his treatment includes neither our enumeration nor the specific order-12 instance — verified against the abstract and secondary literature only, as the *JMT* body is paywalled; the published octatonic commuting groups in this lineage are order 8 ($\cong D_4$, [8]), not order 12. The order-12 instance sits within Peck’s “diagonal cases” remit and is hedged accordingly; the enumeration is the better-supported novelty.

The lineage sidesteps the case (complementary, not missed). Reads of [7, 8, 9] confirm the tritone-central self-dual is unstated: each avoids the reduced-orbit regime differently (ordered pc-segments + a tritone condition; hypothesis-pruning of covers; “no symmetries” + a tritone condition that excludes T_6). Notably [8] tabulates $\{0, 1, 4, 6, 7, 10\}$ (“Major Triad Tritone Mixture”) and studies its 2-triad cover (dual order 2) — the *same set class* as 6-30, never forming the order-12 dual of the hexachord-as-a-point. The proof-assistant literature has only Prismriver [10] (T/I -only, no PLR /duality).

P₆. Where does the tritone lead next?

- **The general- n family.** Proposition 1 is universe-independent; 6-30 is the minimal nontrivial instance of a family (dual $\cong D_{n/2}$ at universe n). A uniform statement and its Lean proof are open.
- **The named isomorphism.** We machine-check the dihedral *fingerprint* (Table 4); the named dual $\cong \text{DihedralGroup } 6$ needs $D_{12}/Z \cong D_6$, currently a Mathlib gap (it is GAP-witnessed here).
- **Fourier.** T_6 -symmetry is exactly even-frequency support of $\hat{A}$ — a magnitude fact; but self-duality needs phase information $|\hat{A}|$ cannot see, the same blindness as the Z-relation. A phase-space account of self-duality is open.
- **Larger chords.** Seventh-chord and tetrachord analogues (cf. [9]): is there a tritone-central self-dual among them?

The tritone that opens the Petrushka chord turns out to be the center of D_{12} , the obstruction in Fourier space, and the seed of an order-12 self-duality — one object seen from the keyboard, the clock, the group, and now the Lean kernel. We have followed it as far as the machine can certify; it points further.

Data and code availability

The Lean sources (`NeoRiemannian.lean`, `SixThirty.lean`) and the Sage/GAP witness scripts of Tables 3–5 accompany this note; each Lean theorem is checkable by `#print axioms` and each script is re-runnable.

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